

Dr. SONU BOSE

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Education

- **Ph.D.:** Department of Mathematics, **Indian Institute of Space Science and Technology, Thiruvananthapuram, Kerala**, 2024.

Date of Open defense and viva-voce examination: December 30, 2024.

Thesis Title: **Efficient Uniformly Accurate Approach to Numerical Solution and Scaled Derivatives for System of Singularly Perturbed Differential Equations.**

- **M.Phil. Mathematics** : University College, **Kerala University**, 2015 (A grade).
- **M.Sc. Mathematics** : Mar Athanasius College, **Mahatma Gandhi University, Kerala**, 2013 (84% marks).
- **B.Sc. Mathematics** : Nirmala college, **Mahatma Gandhi University, Kerala**, 2011 (90% marks).

Post PhD Research Experience

- Senior Project Research Associate, Department of Mathematics, **Indian Institute of Space Science and Technology, Thiruvananthapuram, Kerala**. April 2025 - June 2025
- Research Associate, Department of Mathematics, **Indian Institute of Space Science and Technology, Thiruvananthapuram, Kerala**. December 2024 - March 2025

Broad Research Area

Computational Methods and Analysis of Partial Differential Equations

Research Interests

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| • Approximation of Diffusive Flux for System of ODEs and PDEs | • Higher-Order Approximation using Post-processing Technique |
| • Faster Convergence Analysis on Generalized Adaptive Mesh | • Error Analysis with Optimal-Order Accuracy |
| • Numerical Algorithm for Multi-dimensional System of PDEs with Large Time-lag | |

List of Journal Publications

1. Bose, S., Mukherjee, K. A Parameter-Robust Approach on Generalized Mesh to Numerical Solution and Diffusive Fluxes for 2D-System of Singularly Perturbed Convection-Diffusion Elliptic PDEs, **Applied Numerical Mathematics** [Q1, SCIE/Elsevier], Vol. 226, 1-44, 2026. <https://doi.org/10.1016/j.apnum.2026.02.020>
2. Bose, S., Mukherjee, K. Faster Convergence of Extrapolation Technique for System of Singularly Perturbed Time-Delay Parabolic PDEs, **Mathematical Methods in the Applied Sciences** [Q1, SCIE/Wiley], Vol. 49, 11930-11980, 2026. <https://doi.org/10.1002/mma.70622>
3. Bose, S., Mukherjee, K. Efficient Approximation of Solution Derivatives for System of Singularly Perturbed Time-Dependent Convection-Diffusion PDEs on Shishkin mesh. **Journal of Mathematical Chemistry** [Q2, SCIE/Springer], Vol. 62, 1134-1174, 2024. <https://doi.org/10.1007/s10910-024-01587-8>
4. Bose, S., Mukherjee, K. A fast uniformly accurate global numerical approximation to solution and scaled derivative of system of singularly perturbed problems with multiple diffusion parameters on generalized adaptive mesh. **Computational and Applied Mathematics** [Q2, SCIE/ Springer], 42, 180, 2023. <https://doi.org/10.1007/s40314-023-02218-9>

List of conference Publications

1. Bose, S., Mukherjee, K. Numerical Approximation of System of Singularly Perturbed Convection-Diffusion Problems on Different Layer-Adapted Meshes. B. Das et al. (eds.), *Modeling, Simulation and Optimization*, **Smart Innovation, Systems and Technologies** [SCOPUS/Springer], vol 292, 2022. https://doi.org/10.1007/978-981-19-0836-1_40

Papers Under Preparation

1. Bose, S., Mukherjee, K. Analysis of Diffusive Flux Capturing Approach on Generalized Adaptive Mesh for 2D-System of Singularly Perturbed Elliptic PDEs with Parabolic Boundary Layers, 2026.
2. Bose, S., Mukherjee, K. Crank-Nicolson Based Approximation of Solution Derivatives for System of Singularly Perturbed Time-Delay Parabolic PDEs on Generalized Adaptive Mesh, 2026.

Teaching Experience

☐ Post-PhD Teaching Experience:

- (a) Guest Faculty, Department of Mathematics, **Cochin University of Science and Technology, Kochi**, Kerala. March 2026 - May 2026, July 2026
 - Courses Handled (Current Semester):
 - **Integrated M.Sc.** : Calculus, Linear Algebra
 - **M.Sc.:** Topics in Applied Mathematics
 - Courses Handled (Previous Semester):
 - **Integrated M.Sc.** : Ordinary Differential Equations, Mathematical Methods
 - **M.Sc.:** Partial Differential Equations and Variational Calculus

□ Pre-PhD Teaching Experience:

- (a) Government guest lecturer, Department of Mathematics, Sree Narayana College, Chathannur, **Kerala University**. July, 2017 - January, 2018.
 - Courses Handled:
 - **B.Sc.:** Real Analysis-I, Ordinary Differential Equations
 - **M.Sc.:** Real Analysis
- (b) Guest lecturer, Department of Mathematics, L.B.S. Institute of Technology for Women, Poojapura, **APJ Abdul Kalam Technological University (KTU), Kerala**. August, 2016 - April, 2017.
 - Courses Handled: **B.Tech.** - Ordinary Differential Equations, Calculus
- (c) Guest lecturer, Department of Mathematics, College of Engineering Trivandrum, **APJ Abdul Kalam Technological University (KTU), Kerala**. August, 2015 - January, 2016.
 - Courses Handled:
 - **B.Tech.:** Linear Algebra and Calculus
 - **M.Tech.:** Advanced Mathematics and Optimization Techniques

□ Other Teaching Activities:

- (a) Delivered lectures on *Multivariable calculus* for the internship programme titled " Geometric and Structured of Core Mathematics" at the Department of Mathematics, **Cochin University of Science and Technology, Kochi** in May 2026.
- (b) Delivered tutorial sessions on *Differential Equations* during the Young Talent Nurture (YTN) Programme organized by the Department of Mathematics, **Indian Institute of Space Science and Technology, Thiruvananthapuram, Kerala** in May 2019.

Contributed Talks in Conferences/Symposium

1. **Parameter Uniform Scheme for System of Singularly Perturbed Parabolic Problem with Boundary Layers**. International Conference on Computational Sciences-Modelling Computing and Soft Computing (CSMCS-2020), September 10-12, 2020, Department of Mathematics, NIT, Calicut.
2. **Parameter-uniform convergence analysis for numerical approximation of system of singularly perturbed time-dependent convection-diffusion problems**, international conference on Advances in Differential Equations and Numerical Analysis (ADENA-2020), October 12-14, 2020, Department of Mathematics, IIT, Guwahati.
3. **Parameter-Uniform Numerical Approximation of system of Singularly Perturbed Delay Parabolic PDEs on Generalized Layer-Adapted Mesh**. Symposium on Differential Equations: Analysis, Computation and Applications, Department of Mathematics, IIT, Roorkee, December 2-3, 2021.
4. **Numerical Approximation of System of Singularly Perturbed Convection-Diffusion Problems on Different Layer-Adapted Meshes**, International Conference on Modeling, Simulation and Optimization (CoMSO2021), December 16-18, 2021, Department of Mathematics, NIT, Silcher.

5. **An Efficient Numerical Approximation of System of Singularly Perturbed Time-Delay Convection-Diffusion PDEs on Generalized Adaptive Mesh**, International Conference on Mathematical Sciences and Applications (ICMA-2024), February 15-16, 2024, PG and Research, Department of Mathematics, SH College, Tirupattur, Tamil Nadu.
6. **Parameter-Uniform Numerical Approximation of Singularly Perturbed Elliptic Problems on Different Layer-Adapted Meshes**, 3rd International Conference on Applied Mathematics in Science and Engineering (AMSE-2024), July 25-27, 2024, Centre for data science (CDS) and Department of Mathematics, Siksha 'O' Anusandhan, ITER campus-1, Bhubaneswar.
7. **An Efficient Parameter-Uniform Numerical Approach for Multidimensional Singularly Perturbed Problems**, International Conference on Nonlinear Dynamics: Modelling and Computation (ICNDMC-2025), February 17-19, 2025, Department of Mathematics, Bharathiar University, Coimbatore.
8. **An Efficient Numerical Approximation to Solution and Diffusive Fluxes for 2D-System of Convection-Dominated Elliptic PDEs**, the 2025 UNC Greensboro Virtual Partial Differential Equations Conference, October 10th-12th, 2025.

Workshop/Conference Attended

1. **Instructional School for Teachers on PDE: Theory and Computation**. Department of Mathematics, IISc, Bangalore. 23rd July-04th August, 2018.
2. **SIAM Conference on Analysis of Partial Differential Equations** - Virtual Conference, March 14-18, 2022, Berlin, Germany.

Awards and Achievements

1. Recipient of **Senior Project Research Associate Fellowship**, Indian Institute of Space Science and Technology, Thiruvananthapuram, 2025.
2. Recipient of **Research Associate Fellowship**, Indian Institute of Space Science and Technology, Thiruvananthapuram, 2024.
3. Qualified **Kerala Public Service Commission (KPSC)** Examination for Post of Assistant Professor in Mathematics, 2021.
4. Recipient of **Senior Research Fellowship (CSIR)**, 2020.
5. Recipient of **Junior Research Fellowship (AIR-34) (CSIR)**, 2017.
6. Qualified **NET Lectureship (AIR-30) (CSIR)**, 2016.

Technical Skills

- **Programming language:** MATLAB, C++
- **Familiar language:** Python, Mathematica
- **Software & Tools:** Latex, MS Office

References

1. **Thesis Supervisor:**
Dr. Kaushik Mukherjee,
Professor,
Department of Mathematics,
Indian Institute of Space Science and Technology (IIST), Valiamala P. O., Thiruvananthapuram - 695547,
Kerala, India.
Email: kaushik@iist.ac.in
2. Dr. Prosenjit Das,
Professor,
Department of Mathematics,
Indian Institute of Space Science and Technology (IIST), Valiamala P. O., Thiruvananthapuram - 695547,
Kerala, India.
Email: prosenjit.das@iist.ac.in

Declaration

I certify that the information provided in this resume is true and accurate to the best of my knowledge.